

Labor Market Trajectories and Earnings Dynamics in Colombia

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The Global Repository of Income Dynamics (GRID) Project

- GRID: international database of statistics on income inequality and dynamics.
- Four pillars:
 - **Longitudinal**: follow individuals over time.
 - **Administrative**: reduce measurement error and improve coverage.
 - **Granular**: statistics by age, gender, cohort, and income rank.
 - **Harmonized**: comparable definitions and code across countries.
- Original GRID release: 13 countries; new cohort: 25 countries.

grid-database.org

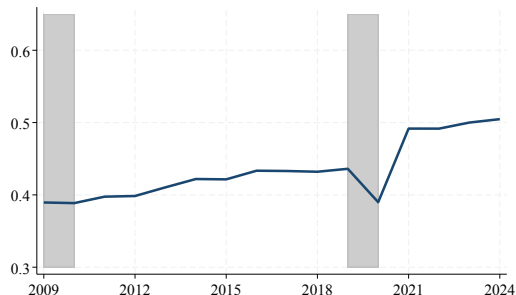
GRID in Colombia

- Extend GRID to Colombia using social security records for *universe of formal workers*.
- Data: *Planilla Integrada de Liquidación de Aportes* (PILA).
- The Colombian setting:
 - formal employment accounts for less than half of the workforce;
 - workers move in and out of formal jobs over their careers;
 - administrative records reveal formal *labor market histories*.

Our contribution: Connect earnings dynamics to long-run histories of *formal attachment*.

Formal Attachment in Colombia

GEIH: Formality Rate



Central labor market challenge: Informality.

- Less than half of workers are formal
- Formality rates are useful snapshots, but miss dynamics.

We provide three sets of new results:

1. Earnings statistics for formal workers
2. Characterize formal attachment histories
3. Connect formal attachment to earnings dynamics

Preview of Results: Earnings Dynamics 2009–2024

- Earnings grew across distribution 2009–2024; concentration declined modestly.
- Earnings dispersion stable; persistently larger below median.
- Formal attachment from labor market histories of 25–30 year olds in 2009
Two thirds of the cohort is either persistently low- or high-attachment.
- Low or falling attachment workers drive earnings dispersion and volatility.
- Formal attachment tied to larger firms, higher earnings, lower earnings risk.

Administrative Data on Formal Workers

PILA: Planilla Integrada de Administración de Aportes

- Administrative registry of mandatory social security contributions.
- Firms and workers report monthly contributions to health and pensions systems.
- The data provide a census of formal employment relationships in Colombia.
- Define a formal worker as someone making social security contributions either as:
 - a wage-earning or salaried employee, or
 - a registered self-employed worker.
- In 2024, 80% of formal workers were employees and 19% were self-employed.

Earnings Measures

- Main outcome: **real annual labor earnings**, y_{it} , following the GRID framework.
 - Nominal earnings deflated to 2018 COP.

- Residualized earnings:

$$\varepsilon_{it} = \log(y_{it}) - \widehat{\log(y_{it})}$$

where fitted component removes age effects, separately by year and sex.

- Permanent earnings: average real earnings over rolling three-year windows.
- Earnings growth:

$$g_{i,t}^h = \varepsilon_{i,t+h} - \varepsilon_{i,t}, \quad h \in \{1, 5\}.$$

Sample Overview

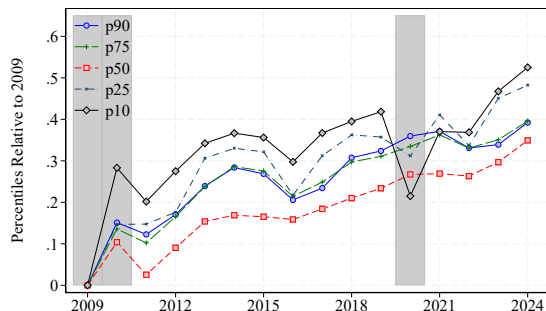
Year		2009	2015	2018	2024
Workers	(M)	7.03	9.39	10.24	11.48
Annual Earnings (\$M 2018 COP)	P10	1.38	2.24	2.34	2.71
	P50	7.61	8.95	9.37	10.82
	P90	17.91	23.08	24.16	27.11
Age Share	25-35	46.5	46.9	46.4	45.7
	36-45	32.7	31.1	32.1	33.5
	45-55	20.7	22.0	21.5	20.8

- PILA coverage grows from 7.0 million formal workers in 2009 to 11.5 million in 2024.
- Earnings percentiles and age composition are computed for workers aged 25 to 55.

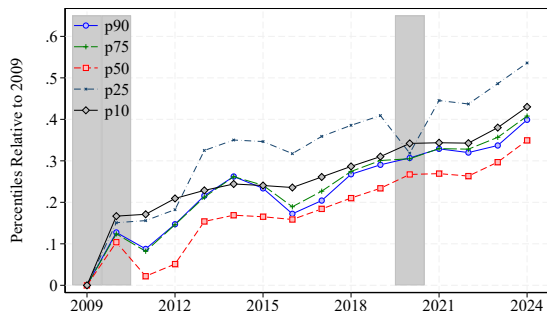
Earnings Inequality

Growth of Earnings Percentiles 2009–2024

Women



Men



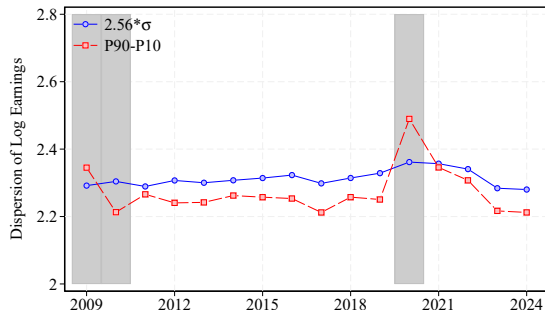
- Growth across the distribution; median 0.3 log points above 2009 levels by 2024.
- Growth inside the top 1% is similar but slightly lower.
- Growth has not meaningfully changed earnings concentration despite modest increase in bottom 50% share.

Top percentiles

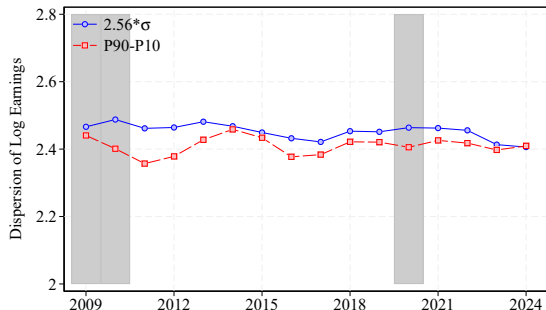
Earnings shares

Earnings Dispersion

Women



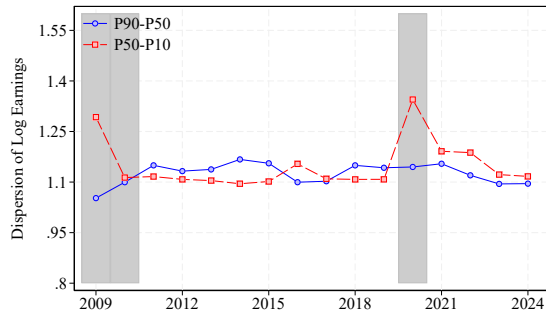
Men



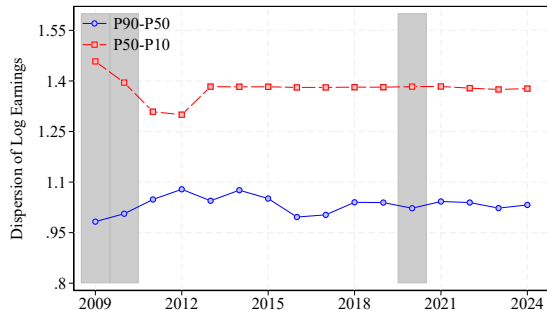
- Earnings dispersion remarkably stable; men's earnings slightly more dispersed.
- The main visible break is a temporary widening of women's earnings distribution during the pandemic.

Earnings Dispersion Above and Below Median

Women



Men

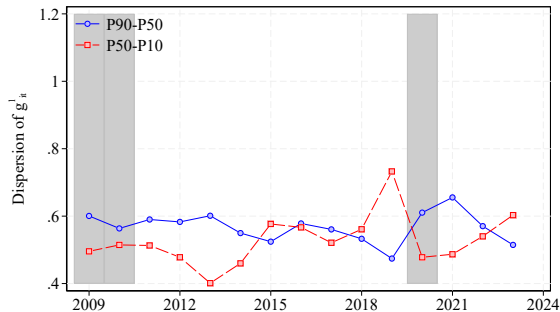


- Dispersion above and below the median is also stable (much higher for market entrants).
- Women's distribution close to symmetric; pandemic spike from lower half.
- Men's distribution is markedly left-skewed (different from other GRID countries).

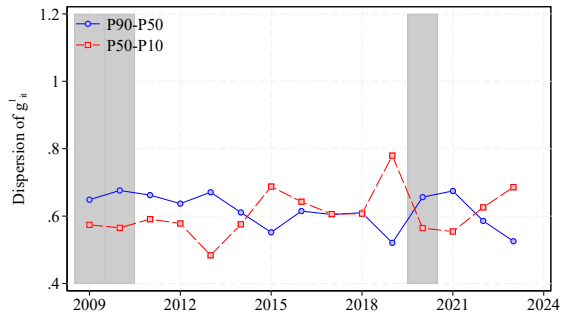
Earnings Growth

(Residualized) Earnings Growth Dispersion

Women



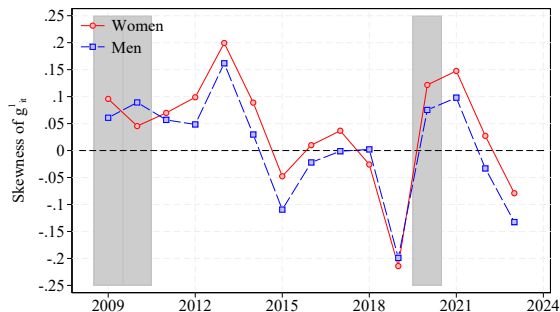
Men



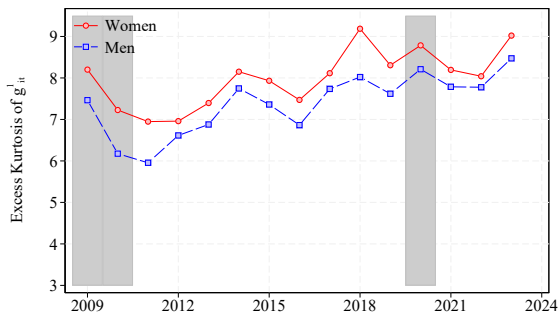
- Distribution is nearly symmetric; women have slightly lower dispersion.
- Below-median growth dispersion rises after 2012.
- The pandemic and recovery periods again make downside growth risk more salient.

Earnings Growth: Skewness and Kurtosis

Skewness



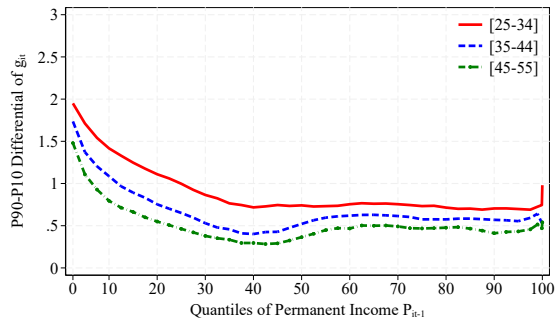
Kurtosis



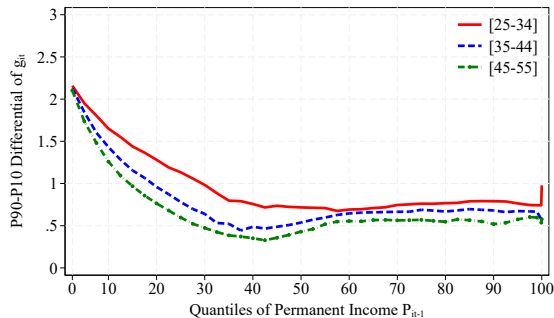
- Skewness turns negative after 2012, dips around pandemic, rebounds afterwards.
- Excess kurtosis is high and growing, consistent with fat-tailed earnings changes.
- Women have consistently higher skewness and kurtosis than men in the sample.

Earnings Growth by Permanent Income

Women



Men



- Earnings growth volatility is largest among young workers and at the bottom of the permanent income distribution.
- Volatility falls through percentile 40 and is flatter through most of the distribution.

Histories of Formal Attachment

Measuring Formal Attachment

- **Objective:** How persistent is formality over work life?
- Track histories of *formal attachment* for a cohort of workers:
 - Workers aged 25–30 in 2009 who appear at least once in PILA between 2009 and 2024.
- Formal attachment for individual i in year t :

$$f_{i,t} = \frac{\text{months with a formal contribution in year } t}{12}.$$

- Each worker is represented by a 16-year vector:

$$F_i = [f_{i,2009}, f_{i,2010}, \dots, f_{i,2024}].$$

- Order matters: formal early and informal later $\neq$ informal early and formal later.

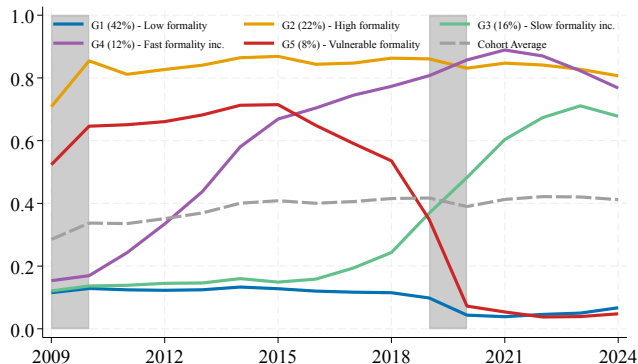
1. Start with each worker as a separate cluster.
2. Merge the two clusters with the closest formal-attachment histories.
3. Replace the merged clusters with their average trajectory.
4. Repeat until all workers are in one group.

Implementation:

$$\arg \min_{g, g'} \sqrt{\frac{2N_g N_{g'}}{N_g + N_{g'}}} \left\| \bar{\mathbf{F}}^g - \bar{\mathbf{F}}^{g'} \right\|_2$$

- Ward's method minimizes within-cluster variance.
- We select **five groups** explaining 60 percent of total variation.

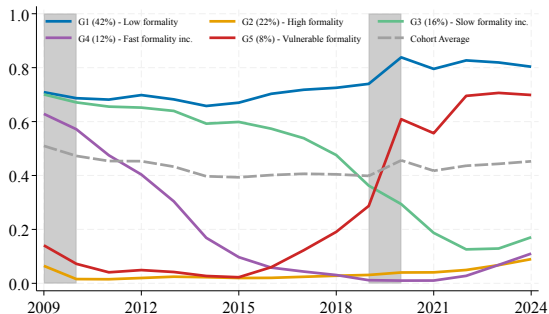
Groups of Formal Attachment



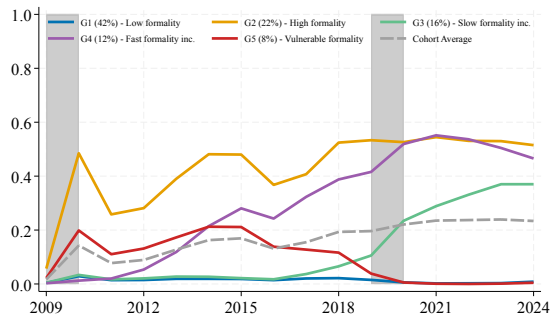
- G1 (42 percent): persistently low attachment (≈ 1 month/year).
- G2 (22 percent): persistently high attachment (≈ 10 month/year).
- G3 and G4 (28 percent): increasing attachment.
- G5 (8 percent): vulnerable attachment.

Extensive Margins of Formal Attachment

No formal contribution



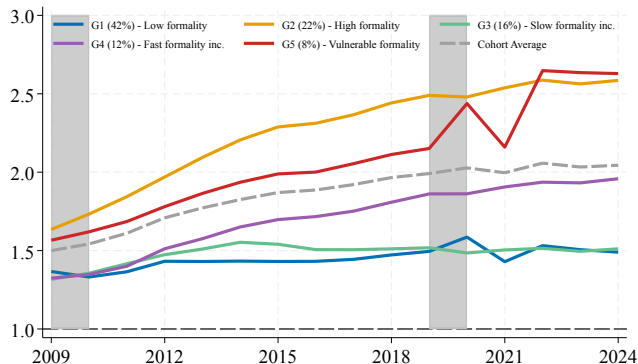
Full-year contribution



- G1 has no formal contribution in most years.
- Even in the high-attachment group, full-year formal employment is rare.

What do histories of formal attachment tell us about earnings?

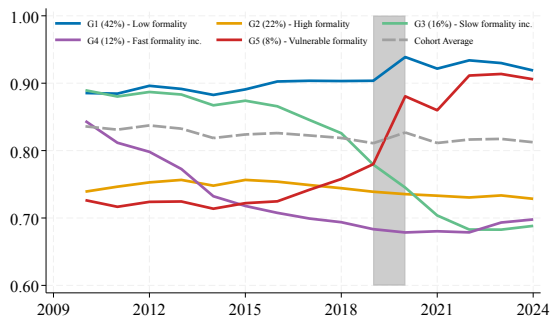
Average Monthly Earnings Relative to the Minimum Wage



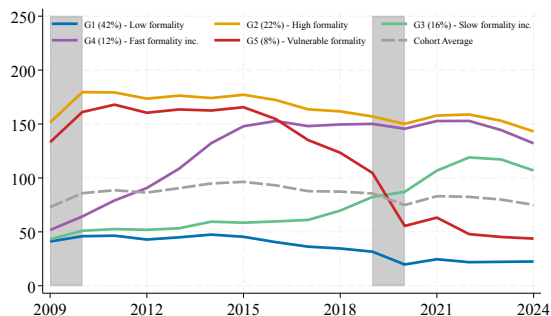
- G2 has the highest formal earnings and the strongest earnings growth.
- G3 becomes more formal but earnings remain low
- G4 increases formality and earnings but does not catch up with G2.
- G5 shows selection: high earners remain formal.

Location and Firm Size

Main metropolitan areas



Average firm size

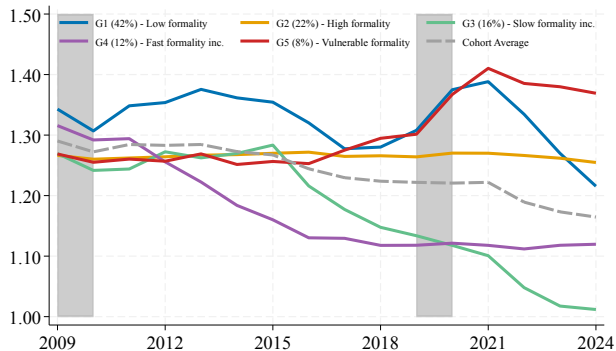


- (Surprisingly?) Formal attachment lower in large cities → Role of public sector.
- High formal attachment is closely linked to larger firms.

Industries

Gender and Formal Attachment

Male-to-Female ratio among formal workers

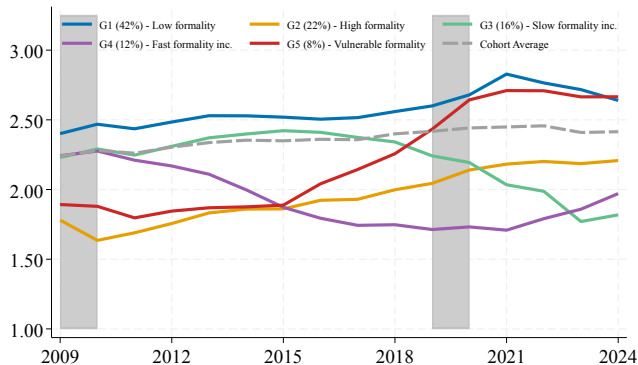


- Formal employment is disproportionately male.
- Increases in formality largely driven by women.
- Pandemic collapse in G5 disproportionately female.

Formal Attachment and Earnings Dynamics

Earnings Dispersion Across Attachment Groups

Standard Deviation Log-Earnings

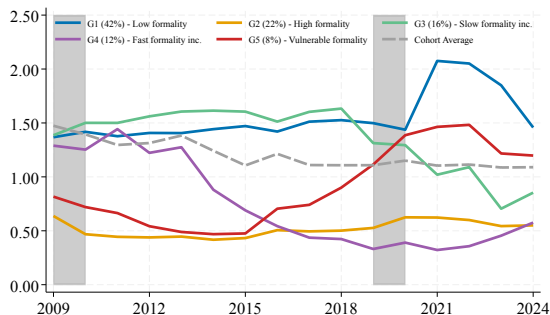


Earnings dispersion driven by workers with low formal attachment.

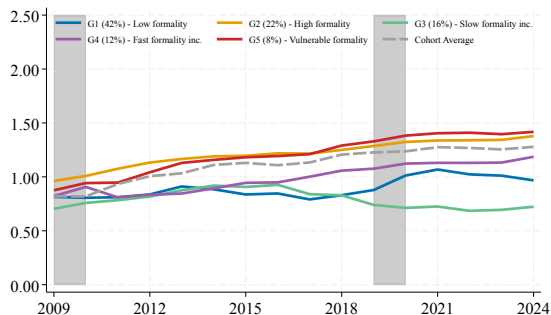
- G1 has the highest dispersion; G2 the lowest.
- G3 and G4 compress as their formal attachment rises.
- G5's falling attachment comes with rising earnings dispersion.

Differences in Earnings Dispersion from Bottom of Distribution

P50-P10



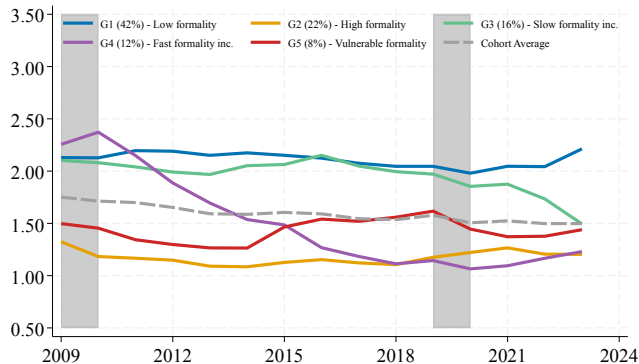
P90-P50



- Earnings dispersion driven by the bottom half of distribution.
- Upper-half dispersion similar across groups; Lower-half dispersion reflects formality.

Earnings Growth Across Attachment Groups

Standard Deviation Log-Earnings Growth

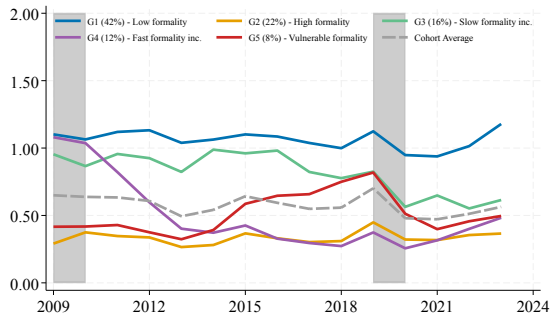


Earnings growth volatility driven by workers with low formal attachment.

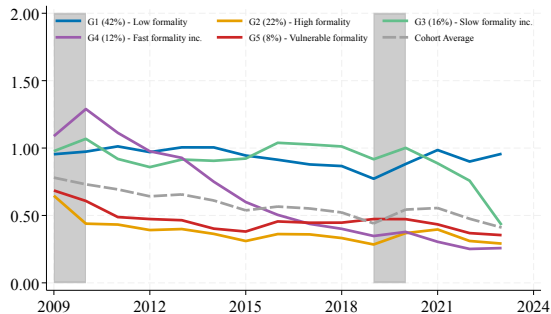
- G1 remains the highest-volatility group throughout the sample.
- G4 shows the sharpest decline in volatility as its attachment to formal work rises.

Upside and Downside Growth Risk Across Attachment Groups

P50-P10



P90-P50



- Similar earnings growth dispersion above and below median (unlike earnings levels).
- Formal attachment negatively correlated with upside and downside growth dispersion.
- Exception is G5: its level dispersion rises as attachment falls, but growth dispersion does not rise as sharply (consistent with selection).

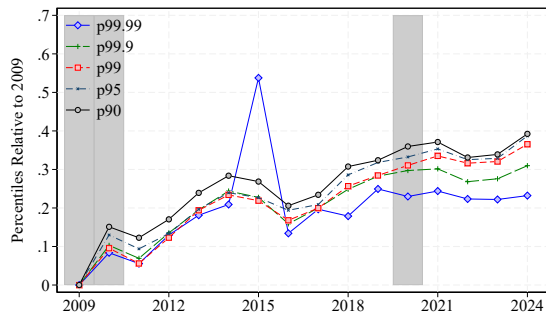
Conclusions

- Colombia's formal workers experienced broad earnings growth and modest declines in concentration from 2009 to 2024.
- Stable aggregate inequality masks sharp differences in formal labor market histories.
- Formal attachment is a persistent and economically meaningful dimension of labor market segmentation.
- Workers with weak attachment drive earnings dispersion and earnings growth volatility.
- For policy, the object is not only the formality rate at a point in time, but the durability of formal jobs over workers' careers.

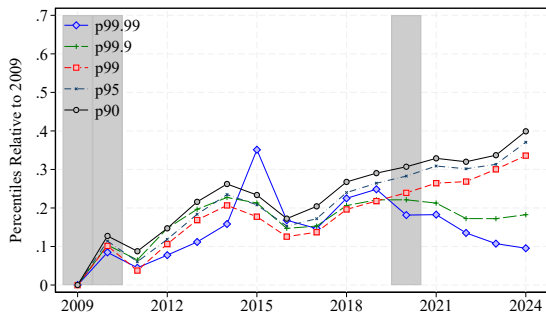
Thank you

Appendix: Growth of Top Earnings Percentiles

Women



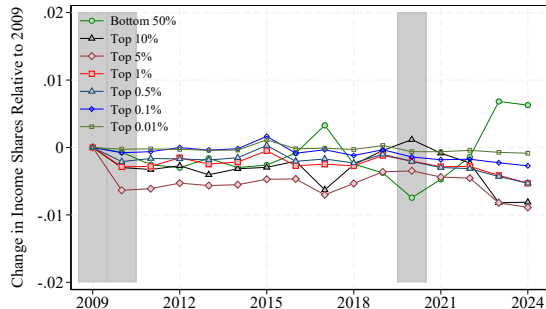
Men



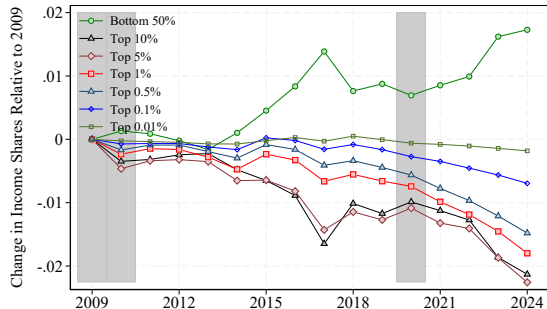
- Growth is stronger around P90 and P95 than at the very top of the distribution.
- The 99.9th and 99.99th percentiles flatten after 2018 for women and decline for men.

Appendix: Earnings Shares

Women



Men



- Earnings concentration declines modestly, especially through gains in the bottom half of the men's distribution.
- Top earnings shares fall more visibly for men; women's earnings shares are comparatively stable over the period.

Appendix: Toy Trajectories

Trajectory	$t = 1$	$t = 2$	$t = 3$	Description
F_1	0.00	0.10	0.20	Low attachment, slight increase
F_2	0.75	0.80	0.85	High attachment
F_3	0.20	0.50	0.85	Increasing attachment
F_4	0.05	0.10	0.20	Low attachment, close to F_1
F_5	0.85	0.80	0.70	High attachment, slight decrease
F_6	0.85	0.45	0.10	Decreasing attachment

- F_1 and F_4 are close: $\|F_1 - F_4\|_2 = 0.050$.
- F_1 and F_2 are far apart: $\|F_1 - F_2\|_2 = 1.214$.

Appendix: First Merge

	F_1	F_2	F_3	F_4	F_5	F_6
F_1	—					
F_2	1.214	—				
F_3	0.789	0.626	—			
F_4	0.050	1.184	0.778	—		
F_5	1.209	0.180	0.731	1.175	—	
F_6	0.925	0.834	0.994	0.879	0.695	—

- Closest pair: F_1 and F_4 .

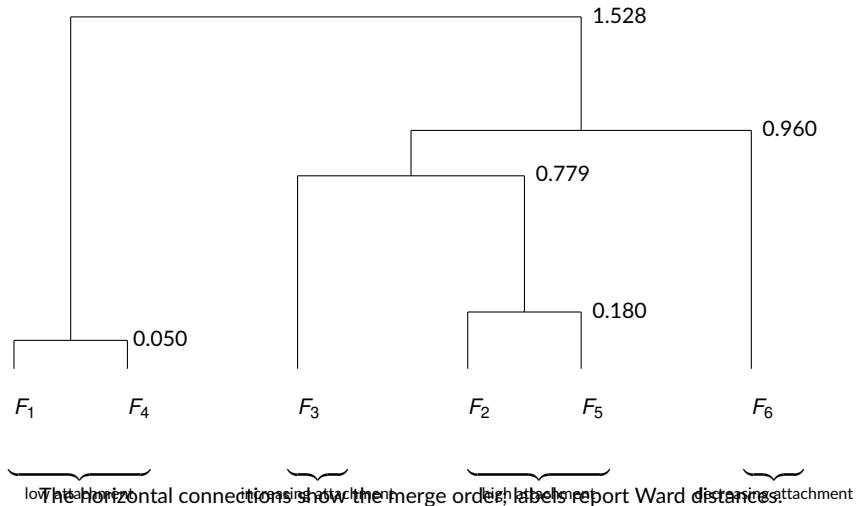
- Merge them into their centroid:

$$\begin{aligned} F_{14} &= (F_1 + F_4)/2 \\ &= (0.025, 0.100, 0.200). \end{aligned}$$

- Next closest pair after updating distances: F_2 and F_5 .

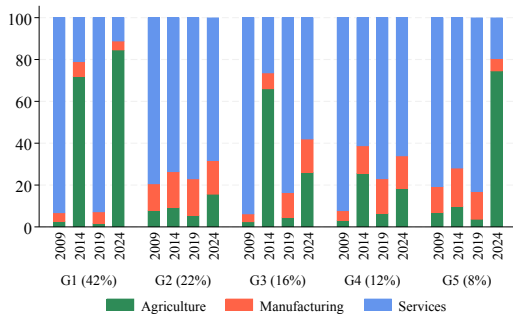
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Appendix: Dendrogram

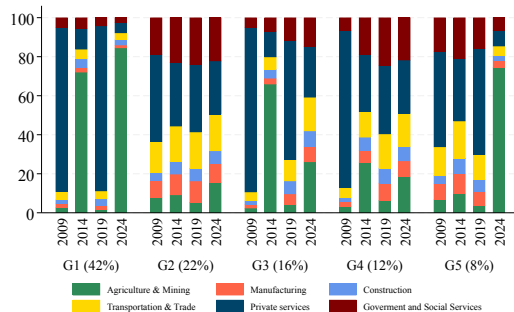


Appendix: Sectors and Industries

Sectors



Industries



- Services dominate formal employment, but sectoral composition varies across attachment groups.
- Government and social services become more prevalent in groups with rising formal attachment.